BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Records and Structured Data

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How is a cell selected in a two-dimensional array?

2. What is a router's main job?

A) Store user passwords
B) Print network documents
C) Run spreadsheet formulas
D) Link different networks

3. "An IPv4 address is always six pairs of hexadecimal digits separated by colons." Correct this. Explain why.

4. Tablets need to join a wired LAN. "A WAP changes domain names into IP addresses." Correct this. How does a WAP help the tablets?

Today's objectives

Group related fields within a record. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

Record

Groups related fields about one item or entity, potentially with different data types.

Field

A field is one named property within a record such as name or age.

Data types

A pupil record might store a string name, integer age and real score.

1D array versus record

An array stores indexed values of one type; records group named fields of potentially different types.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Record	Groups related fields about one item or entity, potentially with different data types. Apply this in a pupil profile.
Field	A field is one named property within a record such as name or age. Apply this in a library book.
Data types	A pupil record might store a string name, integer age and real score. Apply this in a sports member.
1D array versus record	An array stores indexed values of one type; records group named fields of potentially different types. Apply this in a patient appointment.

I Do - One worked example

A pupil profile is being developed. Explain how record and field are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Record [1]; Groups related fields about one item or entity, potentially with different data types. [1].

3. Field [1]; A field is one named property within a record such as name or age. [1].

4. Two linked points for each concept.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A library book uses record. Explain how it helps. [2 marks]

2. Explain field in a sports member. [2 marks]

3. A patient appointment uses data types and 1d array versus record. Explain both. [4 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Each row is a pupil record: column 0 is the name; column 1 is the tutor group. Complete the index to print the second pupil's name.

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "10A"
pupils[1,0] = "Ben"
pupils[1,1] = "10B"
print(pupils[...])
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how field and data types help with a train passenger. [4 marks]

Q4. A programmer designs a school register. Explain why record, data types and 1d array versus record matter. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A record groups related fields about an item, even when the fields have different data types. ☐ T ☐ F
2. Groups related fields about one item or entity, potentially with different data types ☐ T ☐ F
3. A field is one named property within a record such as name or age ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Use two indexes: one for the row and one for the column. ☐

Q2 D - Link different networks ☐

Q3 IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots. ☐

Q4 A wireless access point connects wireless devices to a network. It has a different job from a switch or router. ☐

Code First - Model

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "10A"
pupils[1,0] = "Ben"
pupils[1,1] = "10B"
print(pupils[1,0])
```

Output: Ben

We Do - Check the model

1. Record [1]; Groups related fields about one item or entity, potentially with different data types. [1]. ☐

2. Field [1]; A field is one named property within a record such as name or age. [1]. ☐

3. Data types [1]; A pupil record might store a string name, integer age and real score. [1]. 1D array versus record [1]; An array stores indexed values of one type; records group named fields of potentially different types. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Record [1]; Data types [1]. ☐

Q3 Field [1]; A field is one named property within a record such as name or age. [1]. Data types [1]; A pupil record might store a string name, integer age and real score. [1]. ☐

Q4 Record [1]; Groups related fields about one item or entity, potentially with different data types. [1]. Data types [1]; A pupil record might store a string name, integer age and real score. [1]. 1D array versus record [1]; An array stores indexed values of one type; records group named fields of potentially different types. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: